



Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview from the Industrial Revolution to 2024

Data sources: Global Carbon Budget 2025 · NASA / NOAA Mauna Loa Observatory

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Latest NOAA Mauna Loa reading — February 2026

> 50% increase from pre-industrial baseline of ~280 ppm

Current level is at least 1.5× greater than the natural increase at the end of the last ice age 20,000 years ago.

That ancient change took thousands of years; the modern rise has occurred in roughly two centuries.

Ground-based and satellite measurements confirm the sharp acceleration, linked primarily to burning coal, oil, and natural gas.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

1950

~6 billion tonnes

1990

~20 billion tonnes

2024

>35 billion tonnes

Trajectory description: line chart showing near-zero emissions before 1750, slow growth through the 19th century, steep acceleration after 1950, and continued rise to >35 Bt today.

Reference: [figure_2.jpg](#) — Annual global CO₂ emissions line chart (1750–2024).

Before the Industrial Revolution, CO₂ emissions were negligible.

Growth remained relatively slow until the mid-20th century, then quadrupled by 1990.

While emissions growth has slowed in recent years, they have not yet peaked.

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Country / Region	Approx. Annual CO ₂ (billion t)	Share of Global	Trend (recent)
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Roughly stable
France	~0.3	~0.8%	Declining

Reference: [figure_3.jpg](#) — Annual CO₂ emissions by country.

China is the world's largest emitter by a significant margin — more than one-quarter of the global total.

The United States peaked near 6 billion tonnes around 2005 and has since declined to roughly 5 billion tonnes.

India has risen rapidly to approximately 3 billion tonnes, becoming the third-largest emitter.

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

Visual: stacked area or side-by-side comparison showing regional emission shares from 1900 to 2024.

1900

Europe and the US accounted for more than 90% of global CO₂ emissions.

Asia, Africa, and South America represented a marginal share of the global total.

2024

Asia — home to nearly 60% of the world's population — produces roughly half of all emissions.

The US and Europe combined account for less than one-third.

Africa and South America each contribute only 3–4%, comparable to international aviation and shipping combined.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Reference: figure_1.jpg — Per capita CO₂ emissions over time.



The average American or Australian emits in under two days what the average person in Chad or Niger emits in an entire year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Reference: figure_4.jpg — Annual percentage change in CO₂ emissions, 2024.

Declining (Blue)

- United States: roughly −1% to −5%
- Much of Europe: −1% to −5%
- Some South American countries also recorded declines

Rising (Orange / Red)

- Russia: increases in the positive range
- Parts of Southeast Asia: +2% to +10%
- Several African nations: +2% to +10% or more
- One or two South American countries: very large percentage increases (>10%)

Interpretation: the 2024 map reveals a mixed global picture — mature economies are decoupling growth from emissions, while emerging and developing regions continue to see rising output.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.

Country	Nuclear & Renewables Share	Fossil Fuel Share	Per Capita CO ₂ (t/yr)	Trend
France / UK	High	Lower	~4 tonnes	Declining
Germany / Netherlands	Lower	Higher	Higher than ~4 t	Declining

Key insight: France and the UK have much lower per capita emissions than Germany or the Netherlands, largely because of higher nuclear and renewable shares in their energy mix.

Policy choices — not only prosperity — determine per capita outcomes.

Nuclear and renewable energy deployment can decouple electricity generation from CO₂ emissions at scale.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1. Global CO₂ emissions now exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
2. China alone accounts for over 25% of the global total (~12.5 Bt) — more than double the United States (~5 Bt).
3. Per capita inequalities remain extreme: a ~150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).
4. The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today, while the US and Europe combined are below one-third.
5. Policy and energy mix choices — not just prosperity — determine per capita outcomes. Countries with similar living standards (e.g., France vs. Germany) can have very different emissions profiles.